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Communication Systems II

Seeing Signals: An Interactive RF Constellation Visualiser
and a Synthetic Image Dataset for Automatic Modulation Classification

Abstract

We present an end-to-end pipeline for Automatic Modulation Classification (AMC) built around a custom interactive ground-station visualiser. A Qt/C++ application uses the `liquid-dsp` library to synthesise sixteen digital modulation schemes at complex baseband, corrupts each symbol with a configurable chain of physical impairments (additive white Gaussian noise, oscillator phase noise, I/Q imbalance and a coherent continuous-wave interferer), and renders the resulting constellation as a phosphor-style image. Driven in an automated mode, the same engine exports a labelled dataset of 3072 images (224×224) spanning a five-dimensional sweep of modulation and impairment severity. We analyse each impairment model mathematically, validate the additive-noise channel against the theoretical symbol-error probability of square QAM, and train two image classifiers — a ResNet-18 convolutional network and a LoRA fine-tuned SmolVLM-256M vision-language model — reaching 80% and 55% validation accuracy respectively. The confusion structure exposed a modelling error in our hexagonal-QAM constellations, which we trace and correct. All artefacts are reproducible from the accompanying application.

1 Introduction and project overview

1.1 Aim

Automatic Modulation Classification (AMC) is the task of identifying the modulation scheme of a received radio signal without prior agreement between transmitter and receiver. It is a building block of cognitive radio, spectrum monitoring and electronic warfare, and is increasingly tackled with deep learning applied either to raw I/Q sequences or to image representations of the signal [1]. The aim of this project is to build, from the ground up, the entire chain required to attack AMC as an image-classification problem: a signal generator, a physically-motivated channel model, a rendering/export stage, and finally the learned classifiers.

1.2 How it was achieved

Rather than downloading a ready-made dataset, we implemented an interactive ground-station visualiser as a native Qt 6 / C++ application. The digital-signal-processing core is built on the open-source `liquid-dsp` library [2], which provides the modulator/demodulator (*modem*) objects used throughout. The application runs two pipelines that share the same engine: a *live* mode, in which the user explores the effect of each impairment on the constellation in real time, and an automated *dataset* mode, in which the engine sweeps every modulation/impairment combination and exports a labelled image for each. The full pipeline is therefore self-contained:

bits $\rightarrow$ I/Q symbols $\rightarrow$ impaired constellation $\rightarrow$ labelled image $\rightarrow$ classifier.

1.3 On the choice of models

We make no claim that the two classifiers chosen are optimal; they were selected pragmatically to contrast two paradigms on identical data. The first is a compact, conventional convolutional network (ResNet-18 [3]), the natural baseline for 224×224 images. The second is a small instruction-tuned vision-language model (SmolVLM-256M [4]), fine-tuned with Low-Rank Adaptation (LoRA) [5], to probe whether a general multimodal model can “read” a constellation. Both are small enough to train on modest hardware; a systematic architecture search was out of scope.

2 The signal and data-generation pipeline

This section follows a single symbol through the engine, from bits to a rendered point, and then describes how the labelled dataset is assembled.

2.1 From bits to I/Q symbols

At complex baseband a digitally-modulated signal is a sequence of complex symbols $s = I + jQ$, where the in-phase I and quadrature Q components modulate two orthogonal carriers. Each modem maps $\log_2 M$ input bits to one of M constellation points. We expose the sixteen schemes required by the assignment: amplitude-shift keying (4/8-ASK), phase-shift keying (BPSK, QPSK), square and cross quadrature-amplitude modulation (16/32/64/128/256-QAM), amplitude/phase-shift keying (16/32/64/128-APSK) and hexagonal QAM (4/16/64-HQAM). All built-in modems are normalised to unit average symbol energy, $E[|s|^2] = E_s = 1$, so that the signal-to-noise ratio is well defined and identical across schemes. Figure 1 shows one clean example of each scheme, captured directly from the application.

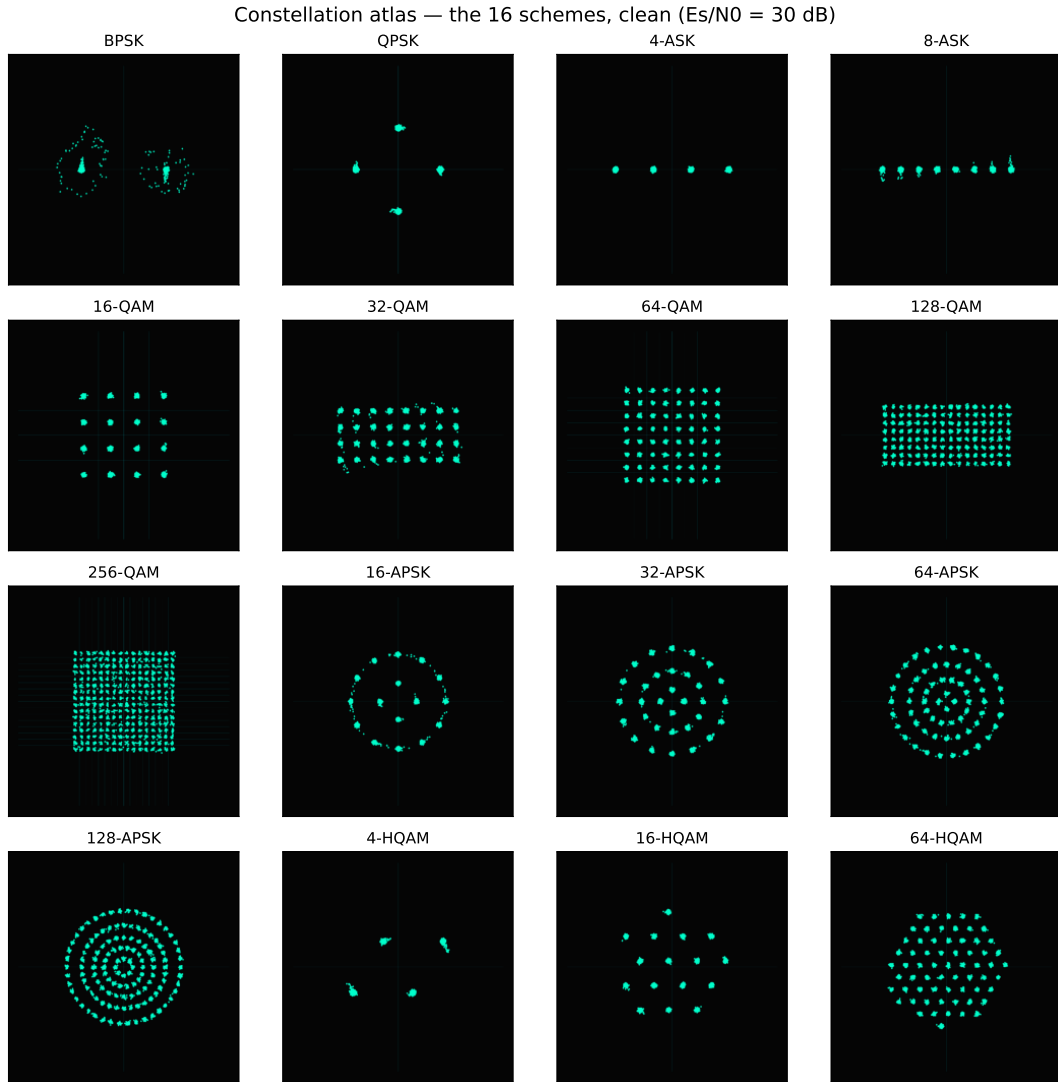


Figure 1: The sixteen modulation schemes rendered by the application at $E_s/N_0 = 30$ dB with no other impairments. The amplitude (ASK), phase (PSK), square (QAM), circular (APSK) and hexagonal (HQAM) geometries are clearly distinct.

2.2 The channel: a chain of physical impairments

The heart of the project is the impairment model. A transmitted symbol is passed through four cascaded effects, applied in the physical order in which they occur along the link: transmitter I/Q imbalance, channel phase rotation, external interference and, finally, receiver thermal noise. Let the ideal symbol be (I, Q) .

(1) I/Q imbalance. Gain and phase mismatch between the in-phase and quadrature branches of the front end is modelled, with the in-phase branch as reference, by

$$\hat{I} = I, \quad \hat{Q} = g(Q \cos \varphi - I \sin \varphi), \quad (1)$$

where g is the relative gain and φ the orthogonality error. This shears and scales the constellation (Figure 2, third row).

(2) Phase noise. A free-running oscillator does not have a perfectly stable phase; its phase performs a random walk (a Wiener process), $\theta[n] = \theta[n-1] + w[n]$ with $w \sim \mathcal{N}(0, \sigma_\Delta^2)$, whose variance grows without bound. In practice a carrier-recovery loop tracks and removes the slow drift, leaving a *bounded residual*. We model this residual as a first-order autoregressive (Ornstein–Uhlenbeck) process,

$$\theta[n] = (1 - \alpha) \theta[n-1] + \sigma_{\text{step}} w[n], \quad w \sim \mathcal{N}(0, 1), \quad (2)$$

whose stationary variance is $\text{Var}(\theta) = \sigma_{\text{step}}^2 / (\alpha(2 - \alpha))$. We therefore set $\sigma_{\text{step}} = \sigma_\theta \sqrt{\alpha(2 - \alpha)}$ so that σ_θ is the target residual RMS phase; α (here 0.02) sets the loop memory ($\sim 1/\alpha$ symbols). The symbol is rotated by $\theta[n]$. Unlike an independent per-symbol jitter, the random walk produces the characteristic rotational smear of Figure 2 (second row) and, crucially, an irreducible error floor (Section 2.4).

(3) Continuous-wave interference. A narrow-band jammer is modelled as a coherent tone of amplitude A rotating at a fixed normalised offset frequency f_{off} : $j[n] = A e^{j2\pi f_{\text{off}} n}$. Because the tone is coherent, it traces a circle around each constellation point over time, turning every point into a ring (Figure 2, fourth row).

(4) Additive white Gaussian noise. Finally, receiver thermal noise is added as circularly-symmetric complex Gaussian noise $n = n_I + jn_Q$ with $n_I, n_Q \sim \mathcal{N}(0, \sigma^2)$ i.i.d. The complex noise power is $N_0 = E[|n|^2] = 2\sigma^2$, hence $E_s/N_0 = 1/(2\sigma^2)$. To make the user-facing SNR control read E_s/N_0 directly we set

$$\sigma = \frac{1}{\sqrt{2}} 10^{-\text{SNR}/20} \quad \implies \quad \frac{E_s}{N_0} = 10^{\text{SNR}/10}. \quad (3)$$

The $1/\sqrt{2}$ factor matters: omitting it (as an earlier version of the engine did) makes the link 3 dB worse than the label, shifting every error-rate curve. Gaussian samples are drawn with the Box–Muller transform. The rows of Figure 2 isolate each impairment on a 16-QAM constellation.

Discrete severity levels. For dataset generation each impairment is quantised to the four qualitative levels *None/Low/Medium/High*, and the SNR to three levels. Table 1 gives the exact parameter behind each label.

2.3 Detection: the “guess”

For every transmitted symbol the engine also performs hard detection with the `liquid-dsp` demodulator, which returns the nearest constellation point by minimum Euclidean distance. Comparing the transmitted and detected symbols yields the instantaneous symbol- and bit-error counts that drive the live telemetry and the error-rate measurements below.

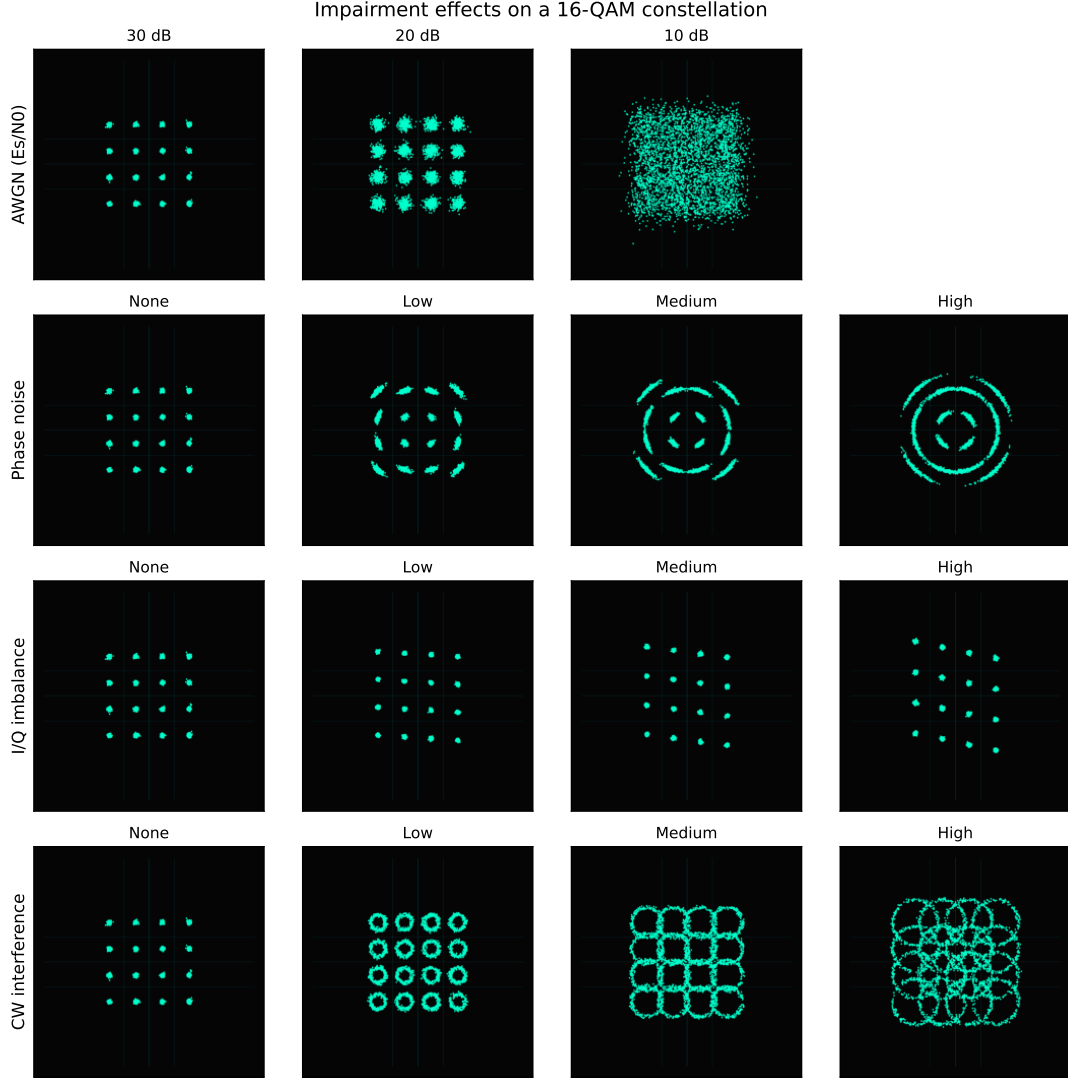


Figure 2: Effect of each impairment on a 16-QAM constellation, captured from the application. Top to bottom: decreasing E_s/N_0 (AWGN cloud); increasing phase noise (rotational smear); increasing I/Q imbalance (shear); increasing CW interference (each point becomes a ring). The faint cross/grid lines are the optional decision-boundary overlay (Section 2.6).

Table 1: Quantitative meaning of the qualitative impairment levels. The SNR uses only the three non-trivial levels; the three impairments additionally include a “None” level.

Severity	E_s/N_0	Phase σ_θ	I/Q (g, φ)	Jammer A
None	—	0°	(1.00, 0°)	0
Low	10 dB	3.4°	(1.06, 3.4°)	0.18
Medium	20 dB	6.9°	(1.12, 6.9°)	0.36
High	30 dB	10.3°	(1.18, 10.3°)	0.54

2.4 Symbol-error probability and the phase-noise floor

To validate the channel model we measure the symbol-error probability (SEP) versus E_s/N_0 for the square constellations 16/64/256-QAM, under both no phase noise and high phase noise, and compare against the textbook expression for square M -QAM [6],

$$P_s = 4\left(1 - \frac{1}{\sqrt{M}}\right)Q\left(\sqrt{\frac{3}{M-1}}\frac{E_s}{N_0}\right) - 4\left(1 - \frac{1}{\sqrt{M}}\right)^2Q^2\left(\sqrt{\frac{3}{M-1}}\frac{E_s}{N_0}\right), \quad (4)$$

with $Q(x) = \frac{1}{2}\text{erfc}(x/\sqrt{2})$. Figure 3 shows the result. With the calibrated noise of (3) the simulated AWGN points fall exactly on the theoretical curves, confirming that the “SNR” control is a true E_s/N_0 . Under high phase noise the curves no longer fall off indefinitely but flatten into an *error floor*: beyond a certain SNR the residual phase, not the thermal noise, dominates, and adding power no longer helps. The floor is highest for 256-QAM and lowest for 16-QAM, since denser constellations have less angular margin — a textbook qualitative result reproduced quantitatively by our engine.

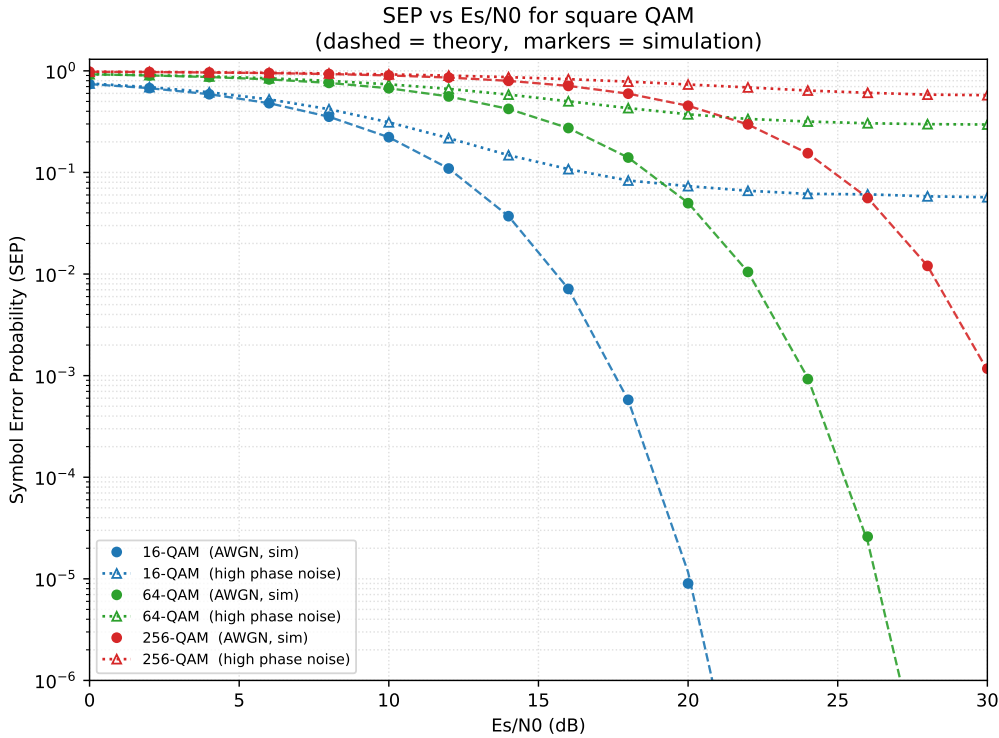


Figure 3: Symbol-error probability versus E_s/N_0 . Markers are the simulation; dashed lines are the theoretical square-QAM SEP. The AWGN-only points match theory; the high-phase-noise curves exhibit an irreducible error floor that worsens with modulation order.

2.5 A corrected modelling error: hexagonal QAM

An early version of the engine mapped the three hexagonal-QAM schemes (4/16/64-HQAM) onto the *square* QAM modems of the same order, because `liquid-dsp` has no built-in hexagonal constellation. This produced images pixel-identical to ordinary QAM and, as Section 3 shows, the classifier could not separate the two — which is how we caught the bug. We corrected it by building true hexagonal constellations on the A_2 lattice (basis $e_1 = (1, 0)$, $e_2 = (\frac{1}{2}, \frac{\sqrt{3}}{2})$): we take the M lowest-energy lattice points, centre them, normalise to unit average energy, and load them into `liquid-dsp`’s arbitrary-constellation modem. The resulting constellations (Figure 4)

are now geometrically distinct from square QAM. We verified numerically that each table is zero-mean, unit-energy, and demodulates losslessly in the absence of noise.

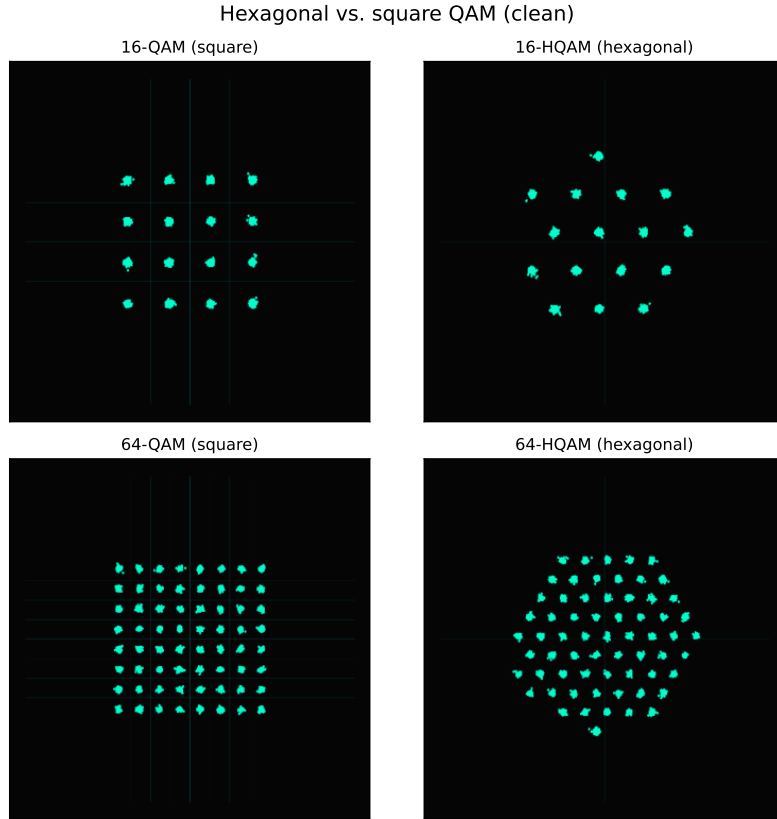


Figure 4: The corrected hexagonal QAM (right column) versus ordinary square QAM (left column). The hexagonal packing is now visible, removing the aliasing that previously made the two indistinguishable.

2.6 Dataset construction

In dataset mode the engine performs a full five-dimensional sweep: 16 modulations, 3 SNR levels, and 4 levels each of phase noise, I/Q imbalance and interference, giving $16 \times 3 \times 4 \times 4 \times 4 = 3072$ configurations. For each, the constellation accumulates a dense cloud of about 3300 points and is captured as a clean 224×224 image (cyan points on black, no axes or labels — the native input size of ResNet/VGG-style networks), together with a CSV row recording the modulation and the four severity labels. A note on reproducibility: in the run used for the figures here the optional decision-boundary overlay was left enabled at low opacity, so faint cross/grid lines are visible on the square-QAM panels; for clean training data this overlay should be disabled (or the application set to skip it during automation). The export takes a few minutes, the engine having been optimised to draw a full image roughly four times faster than the initial implementation.

3 Automatic modulation classification

3.1 Models and training

Both classifiers are trained on the same 3072-image dataset with an 80/20 train/validation split. The convolutional baseline is a ResNet-18 [3] trained with the Adam optimiser and cross-entropy loss for 10 epochs. The vision-language model is SmolVLM-256M-Instruct [4], adapted to the

sixteen-class problem by Low-Rank Adaptation (LoRA [5]; rank 16, $\alpha = 32$, dropout 0.05 on the attention and MLP projections), trained for 3 epochs at learning rate 5×10^{-5} ; only the small adapter is updated, leaving the 256M backbone frozen.

3.2 Convolutional baseline

The ResNet-18 reaches 80% validation accuracy (macro-F1 0.78). Its confusion matrix (Figure 5) splits the errors into two clear families:

- **Hexagonal vs. square QAM aliasing.** 16-HQAM and 64-HQAM have low precision and bleed into 16-QAM and 64-APSK. This is *not* a weakness of the network: in the dataset used for training, HQAM was still rendered as square QAM (Section 2.5), so the two classes were literally the same images and could only be guessed at chance. This failure is precisely what led us to find and fix the constellation bug; with the corrected, geometrically distinct constellations (Figure 4) these classes become separable.
- **High-order density saturation.** 128-APSK (13% recall) and 128-QAM (21% recall) are the densest constellations; 128-QAM is most often confused with 32-QAM. At 224×224 with noise, constellations of 128–256 points blur into near-identical clouds. This is a genuine information limit of the image representation rather than a modelling error, and would require higher resolution or per-class SNR control to mitigate.

The low-order and structurally distinctive schemes (BPSK, QPSK, 4-HQAM, 16/64/256-QAM) are classified almost perfectly.

3.3 Vision-language model

The LoRA-tuned SmolVLM-256M reaches 55% accuracy (macro-F1 0.54). It classifies the visually distinctive classes well (256-QAM, 64-QAM, 16-QAM and QPSK all above 0.84 F1) but struggles far more than the CNN on the APSK family and the high-order schemes (Figure 6). This is the expected outcome: a 256M general-purpose multimodal model, adapted with a small LoRA on a few thousand images, has neither the inductive bias of a convolutional network for dense-texture discrimination nor enough task data to acquire it. The experiment is nonetheless informative — it shows that a general “image reader” can recover the coarse structure of a constellation but not the fine lattice distinctions that separate, say, 64-APSK from 64-HQAM.

3.4 Discussion

On this dataset the convolutional baseline clearly outperforms the fine-tuned VLM (80% vs. 55%), which is unsurprising: distinguishing modulations from constellation images is a fine-grained texture/geometry problem, exactly what convolutional inductive biases are good at, and the VLM’s language head adds little. The most valuable scientific outcome, however, came from the errors rather than the accuracy: the CNN’s inability to separate HQAM from QAM was a direct, visible symptom of a flaw in our signal generator, which we then corrected at the source.

4 The application

The visualiser is a native Qt 6 application with a C++/QML architecture. The DSP engine runs on a dedicated worker thread so the user interface stays responsive, and exposes its parameters to the QML front end through Qt’s property and signal system. The front end provides a live constellation page (a phosphor-persistence canvas that imitates an analogue oscilloscope), a telemetry dashboard (RSSI/SNR/BER/PLR time series), sliders for every impairment, a serial-port interface for connecting a real radio, and the one-click dataset and SEP-curve exporters

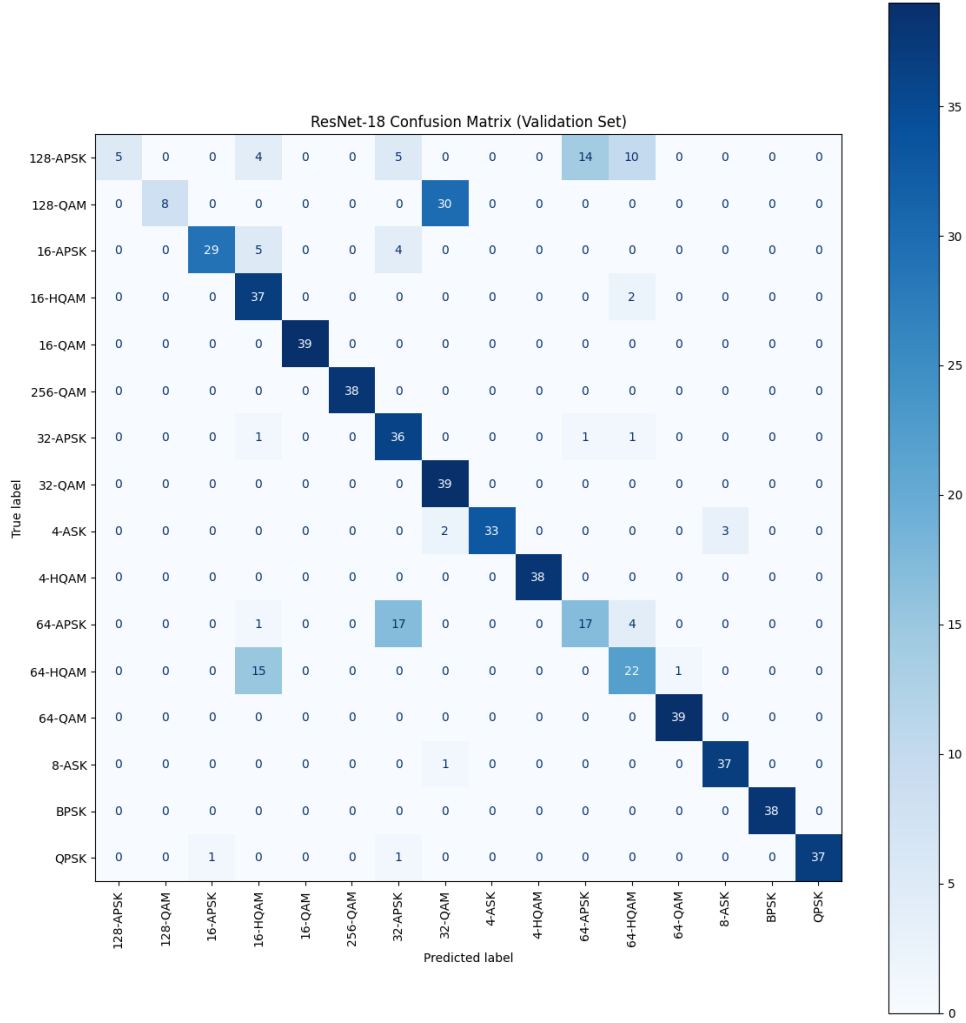


Figure 5: ResNet-18 validation confusion matrix. Errors concentrate in (i) the HQAM/QAM aliasing (since fixed) and (ii) the densest high-order constellations.

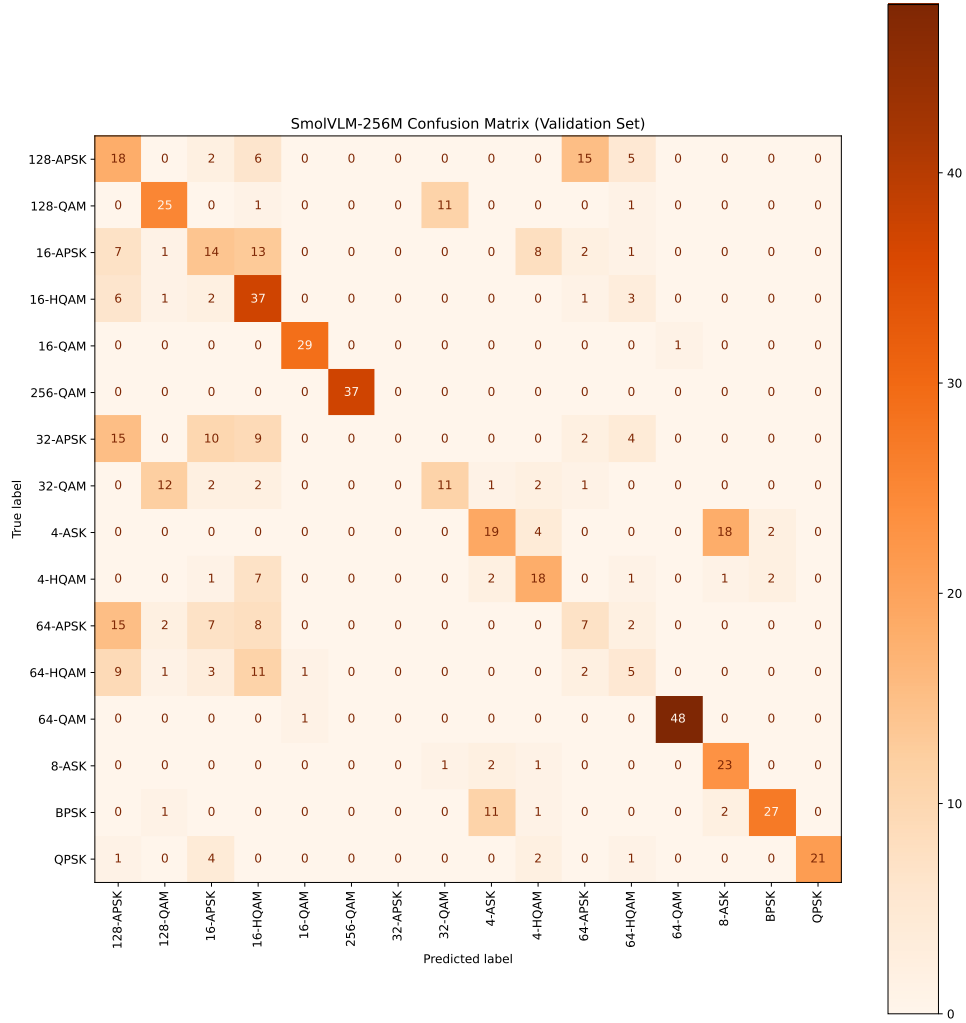


Figure 6: SmolVLM-256M (LoRA) validation confusion matrix. The model captures coarse constellation structure but confuses the dense APSK and high-order classes.

used to produce the data in this report. The application is thus simultaneously a teaching instrument, a data-generation rig, and the experimental apparatus behind every figure here.

5 Conclusion

We built a complete, self-contained AMC pipeline: a Qt/C++ visualiser that synthesises sixteen modulations, corrupts them with a mathematically-grounded chain of impairments (calibrated AWGN, Wiener/Ornstein–Uhlenbeck phase noise, I/Q imbalance and a coherent CW jammer), and exports a labelled 3072-image dataset, on which a ResNet-18 and a LoRA-tuned SmolVLM reach 80% and 55% accuracy. The additive-noise channel was validated against theory, the phase-noise error floor was reproduced, and a constellation-modelling error revealed by the classifier was traced and corrected. Natural extensions include retraining on the corrected hexagonal constellations, adding multipath fading and carrier-frequency offset to the channel, and raising the image resolution to relieve the high-order density saturation.

References

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